FUN-3.B.1 Derivatives of products of differentiable functions can be found using the product rule.
FUN-3.B.2 Derivatives of quotients of differentiable functions can be found using the quotient rule.

2-8 & 2-9 Product and Quotient Rules

Get Ready...

Recall: If $f(x) = 5g(x) - 3$ then $f'(x) = 5g'(x) - 0 = 5g'(x)$

But what if we had $f(x) = g(x)h(x)$...could we find $f'(x)$?
Is $f'(x) = g'(x)h'(x)$?

Let $f(x) = (3x^2 - 2)(4x + 7)$. Find $f'(x)$.

Method #1	Try $f'(x) = g'(x)h'(x)$	$f'(x) = (6x)(4) = 24x$
Method #2	Try to distribute first. Then find $f'(x)$.	$f(x) = 12x^3 + 21x^2 - 8x - 14$ $f'(x) = 36x^2 + 42x - 8$

Since we arrived at different answers for $f'(x)$, the methods cannot both be correct. Which method do you think is right, if any?

Method __2__ was correct. But this method does not always work because...

As a result, it is very useful and convenient to have a rule that can be used when working with products.

The Product Rule	If f and g are both differentiable, then $$\frac{d}{dx}[\underbrace{f(x)}_{\text{first}} \cdot \underbrace{g(x)}_{\text{second}}] = f'(x)g(x) + f(x)g'(x)$$

An easier way to remember this is to shorten the expression:

$$\frac{d}{dx}[f \cdot g] = f'g + fg'$$

Example 1: $f(x) = \underbrace{x^2}_{\text{first}}\underbrace{\sin(x)}_{\text{second}}$. Find $f'(x)$.

$$f'(x) = 2x\sin x + x^2\cos x$$

Example 2: $f(x) = x - 2e^x \cos(x)$. Find $f'(x)$.

$$f'(x) = 1 - 2e^x \cdot \cos x + -2e^x(-\sin x)$$
$$= 1 - 2e^x \cos x + 2e^x \sin x$$

Example 3: $h(x) = \cos(x)\sqrt{x}$. Find $h'(x)$.

$x^{1/2}$

$$h'(x) = -\sin x \cdot \sqrt{x} + \cos x \cdot \frac{1}{2}x^{-1/2}$$

$$= -\sin x\sqrt{x} + \frac{1}{2}\cos x \cdot x^{-1/2}$$

OR

$$-\sin x\sqrt{x} + \frac{\cos x \cdot x^{-1/2}}{2}$$

AP mult choice $= -\sin x\sqrt{x} + \dfrac{\cos x}{2\sqrt{x}}$

Just as some functions are a product of two functions, we will also work with a quotient of two functions. As a result, it is convenient to have a derivative rule for these cases.

The Quotient Rule	If f and g are both differentiable, then $$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

An easier way to remember this is to shorten the expression:

$$\frac{d}{dx}\left[\frac{f}{g}\right] = \frac{gf' - fg'}{g^2}$$

Example 4: Find $\frac{dy}{dx}$ if $y = \frac{5x-2}{x^2+1}$

$$\frac{dy}{dx} = \frac{(x^2+1)(5) - (5x-2)(2x)}{(x^2+1)^2}$$

$$= \frac{5x^2+5+(-10x^2+4x)}{(x^2+1)^2}$$

$$= \frac{-5x^2+4x+5}{(x^2+1)^2}$$

Example 5: Let $f(x) = \frac{3x^2-2x+3}{7}$. Find $f'(2)$.

$\frac{2x}{7} = \frac{2}{7}x$

$f(x) = \frac{3x^2}{7} - \frac{2x}{7} + \frac{3}{7}$

$f'(x) = \frac{6x}{7} - \frac{2}{7}$

$f'(2) = \frac{6(2)}{7} - \frac{2}{7} = \frac{12}{7} - \frac{2}{7} = \frac{10}{7}$

Example 6: Let $g(x) = \frac{\sin(x)}{3x^2}$. Find the instantaneous rate of change of g at $x = 4$. Find $g'(4)$

$$g'(x) = \frac{3x^2(\cos x) - \sin x(6x)}{(3x^2)^2}$$

$$g'(4) = \frac{3(4)^2(\cos 4) - \sin 4(6 \cdot 4)}{(3(4)^2)^2}$$

$$= \frac{48\cos 4 - 24\sin 4}{48^2}$$

M.C Quest: $= \frac{48\cos 4 - 24\sin 4}{48 \cdot 48} \cdot \frac{\frac{1}{24}}{\frac{1}{24}}$

$$= \frac{2\cos 4 - \sin 4}{96}$$

$48 \cdot \cancel{48}^{2} \cdot \frac{1}{\cancel{24}}$

Selected values of the twice differentiable function $f(x)$ are given in the table above along with $f'(x)$ and $f''(x)$. Use this information to find the following.

x	-3	0	1	5
$f(x)$	2	-4	6	3
$f'(x)$	0	5	-2	7
$f''(x)$	9	3	4	2

Example 7: $k(x) = 3\sin(x)f(x)$. Find $k'(0)$.

$k'(x) = 3\cos x \cdot f(x) + 3\sin x \cdot f'(x)$

$k'(0) = 3\cos 0 \cdot f(0) + 3\sin 0 \cdot f'(0)$

$= 3(-4) + 0$

$= -12$

Selected values of the twice differentiable function $f(x)$ are given in the table above along with $f'(x)$ and $f''(x)$. Use this information to find the following.

x	-3	0	1	5
$f(x)$	2	-4	6	3
$f'(x)$	0	5	-2	7
$f''(x)$	9	3	4	2

Example 8: $p(x) = e^x f(x) - f'(x)(x^2 - x)$. Find $p'(5)$.

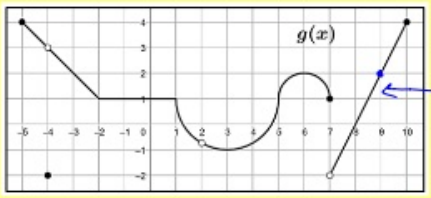
$$p'(x) = e^x \cdot f(x) + e^x \cdot f'(x) - f''(x)(x^2 - x) - f'(x)(2x - 1)$$

$$p'(5) = e^5 f(5) + e^5 f'(5) - f''(5)(20) - f'(5)(9)$$

$$= 3e^5 + 7e^5 - 40 - 63$$

$$= 10e^5 - 103$$

Example 9: The graph of $g(x)$ is shown above and consists of several linear pieces and two semicircles. Use the graph of $g(x)$ to answer the following.



slope

$g'(9) = 2$

a.) $p(x) = g(x)\sqrt{x}$. Find $p'(9)$.

$\sqrt{x} = x^{1/2}$

$9^{-1/2} = \frac{1}{9^{1/2}} = \frac{1}{3}$

$$p'(x) = g'(x)\cdot\sqrt{x} + g(x)\cdot\frac{1}{2}x^{-1/2}$$

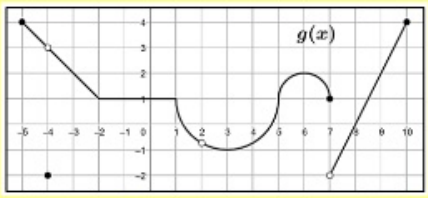
$$p'(9) = g'(9)\cdot 3 + g(9)\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)$$

$$= 2(3) + 2\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)$$

$$= 6 + \frac{1}{3}$$

$$= \frac{18}{3} + \frac{1}{3} = \frac{19}{3}$$

Example 9: The graph of $g(x)$ is shown above and consists of several linear pieces and two semicircles. Use the graph of $g(x)$ to answer the following.



b.) $y = e^x g(x)$. Find $\frac{dy}{dx}$ at $x = 0$.

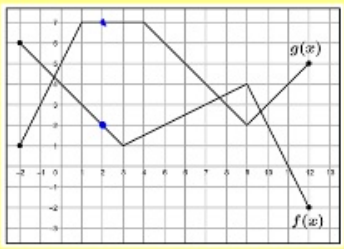
$$\frac{dy}{dx} = e^x \cdot g(x) + e^x g'(x)$$

$$\left.\frac{dy}{dx}\right|_{x=0} = e^0 g(0) + e^0 g'(0)$$

$$= 1(1) + 1(0)$$

$$= 1$$

Example 10: The graphs of $f(x)$ and $g(x)$ are shown above and consist of several line segments. Use the graphs above to find the following, if possible.



a) Let $j(x) = f(x)g(x)$. Find $j'(2)$ and $j'(7)$.

b) Let $j(x) = f(x)g(x)$. Write an equation of the line tangent to j at $x = 7$.

a) $j'(x) = f'(x)g(x) + f(x)g'(x)$

$j'(2) = f'(2)g(2) + f(2)g'(2)$

$= -1(7) + 2(0) = -7$

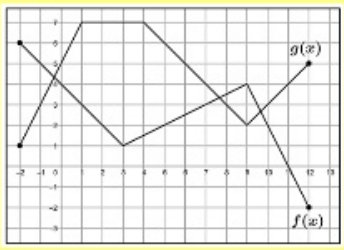
$j'(7) = f'(7)g(7) + f(7)g'(7)$

$= \frac{1}{2}(4) + 3(-1) = 2 - 3 = -1$

b) slope $= j'(7) = -1$ $\quad j(7) = f(7)g(7) = 3(4) = 12$

$y - 12 = -1(x - 7)$

Example 10: The graphs of $f(x)$ and $g(x)$ are shown above and consist of several line segments. Use the graphs above to find the following, if possible.



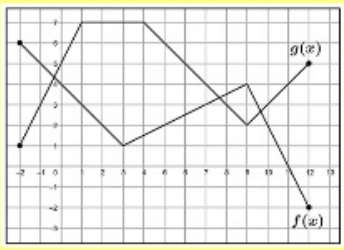
c) Let $h(x) = \dfrac{f(x)}{g(x)}$. Find $h'(0)$.

$$h'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

$$h'(0) = \frac{g(0)f'(0) - f(0)g'(0)}{[g(0)]^2} = \frac{5(-1)-4(2)}{[5]^2}$$

$$= \frac{-5-8}{25} = \frac{-13}{25}$$

Example 10: The graphs of $f(x)$ and $g(x)$ are shown above and consist of several line segments. Use the graphs above to find the following, if possible.



d) Let $w(x) = \dfrac{x^2}{g(x)}$. Find $w'(2)$.

$$w'(x) = \frac{g(x) \cdot 2x - x^2(g'(x))}{[g(x)]^2}$$

$$w'(2) = \frac{g(2) \cdot 4 - 4 \cdot g'(2)}{[g(2)]^2} = \frac{7(4) - 4(0)}{49}$$

$$= \frac{28}{49} = \frac{4}{7}$$